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## 12. Dimension of an eigenspace

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**Theorem 12.1** Let  $\lambda$  be an eigenvalue of an  $n \times n$  matrix  $A$ . Suppose that the multiplicity of  $\lambda$  (as a root of the characteristic polynomial) is  $m$ . Then

$$1 \leq (\text{dimension of eigenspace of } \lambda) \leq m.$$

Given an eigenvalue  $\lambda$ , the dimension of the eigenspace of  $\lambda$  is the maximum number of linearly independent eigenvectors corresponding to  $\lambda$ . This dimension is at least  $1$ , since there is at least one eigenvector of eigenvalue  $\lambda$  (otherwise  $\lambda$  would not have been an eigenvalue). That this dimension is at most  $m$  requires more work to prove and we will not prove this in this course. One of the main ingredients is that eigenvectors corresponding to different eigenvalues must be linearly independent, which we will show below.

**Example 12.2** A  $9 \times 9$  matrix has characteristic polynomial  $(\lambda - 2)^3(\lambda - 5)^6$ . What are the possibilities for the dimension of the eigenspace of  $2$ ?

**Solution:** In this case,  $m = 3$ , so the dimension is  $1, 2$ , or  $3$ .

**Definition 12.3**

- The eigenvalue  $\lambda$  and its eigenspace are called **complete** (or **perfect**) if the dimension of the eigenspace equals the multiplicity  $m$  of  $\lambda$ .
- The eigenvalue  $\lambda$  and its eigenspace are called **deficient** (**defective**, **degenerate**, or **incomplete**) if the dimension of the eigenspace is less than  $m$ .
- A **matrix** is **complete** if **all** of its eigenspaces are complete; it is **deficient** if any **one** of its eigenspaces is deficient.

**Warning :** Different authors use different terminology here.

**Example 12.4** If the multiplicity of an eigenvalue is  $1$ , then the dimension of its eigenspace is sandwiched between  $1$  and  $1$ , and so must be  $1$ . In other words, the eigenspace of any eigenvalue of multiplicity  $1$  is complete.

**Theorem 12.5** Fix an  $n \times n$  matrix  $A$ . The eigenvectors corresponding to distinct eigenvalues are linearly independent.

proof

Suppose that  $\mathbf{v}_1, \dots, \mathbf{v}_p$  are eigenvectors corresponding to distinct eigenvalues  $\lambda_1, \dots, \lambda_p$ . Suppose that there were a linear relation

$$c_1 \mathbf{v}_1 + \dots + c_p \mathbf{v}_p = \mathbf{0}.$$

Apply  $\lambda_1 \mathbf{I} - A$  to both sides; this gets rid of the first summand on the left. Next apply  $\lambda_2 \mathbf{I} - A$ , and so on, up to  $\lambda_{p-1} \mathbf{I} - A$ . This shows that some nonzero number times  $c_p \mathbf{v}_p$  equals  $\mathbf{0}$ . But  $\mathbf{v}_p \neq \mathbf{0}$ , so  $c_p = 0$ . Similarly each  $c_i$  must be  $0$ . Thus only the trivial relation between  $\mathbf{v}_1, \dots, \mathbf{v}_p$  exists, so they are linearly independent.

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As a consequence,

**Fact:** If all the eigenspaces of a  $n \times n$  matrix  $A$  are complete, then there is a set of  $n$  linearly independent eigenvectors of  $A$ .

proof

Choose a basis for each eigenspace and concatenate these lists of eigenvectors. Since all the eigenspaces are complete, the number of linearly independent eigenvectors from each eigenspace is the multiplicity of the eigenvalue. So the total number of eigenvectors in our list is the sum of the multiplicities of the eigenvalues, which is  $n$ , the degree of the characteristic polynomial.

The theorem above tells us that eigenvectors of different eigenvalues are linearly independent, so our list consists of all linearly independent eigenvectors. (This list forms a basis of  $\mathbb{C}^n$  or  $\mathbb{R}^n$ ).

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**Fact:** If instead the matrix  $A$  is deficient, then the maximum number of linearly independent eigenvectors is less than  $n$ .

**Example 12.6** Recall for the matrix  $\begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 1 \end{pmatrix}$ , the eigenvalues and corresponding eigenvectors are :

$$\lambda = 0; \begin{pmatrix} 1 \\ 0 \\ -1 \end{pmatrix} \tag{3.25}$$

$$\lambda = 1; \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} \tag{3.26}$$

$$\lambda = 2; \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix}. \tag{3.27}$$

There are 3 distinct eigenvalues (each of multiplicity **1**). So the corresponding eigenvectors  $\begin{pmatrix} 1 \\ 0 \\ -1 \end{pmatrix}$ ,  $\begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}$ , and  $\begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix}$  are linearly independent and form a basis of  $\mathbb{R}^3$  (or  $\mathbb{C}^3$ ).

**Example 12.7** For  $\begin{pmatrix} -2 & 1 & 1 \\ 1 & -2 & 1 \\ 1 & 1 & -2 \end{pmatrix}$ , the eigenvalues and corresponding eigenvectors are

$$\text{Eigenvalue} \quad \text{Eigenvectors (basis of eigenspace)} \tag{3.28}$$

$$\lambda = 0; \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} \tag{3.29}$$

$$\lambda = -3; \begin{pmatrix} -1 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} -1 \\ 0 \\ 1 \end{pmatrix} \tag{3.30}$$

The eigenvalue  $-3$  has multiplicity **2** (as a root of the characteristic polynomial) and its eigenspace is of dimension **2** and has a basis  $\begin{pmatrix} -1 \\ 1 \\ 0 \end{pmatrix}$ ,  $\begin{pmatrix} -1 \\ 0 \\ 1 \end{pmatrix}$ . These two eigenvectors of  $-3$  together with any one eigenvector of the eigenvalue **0** form a basis of  $\mathbb{R}^3$  since the 3 vectors must be linearly independent.

### A deficient example



This possibility of a repeated eigenvalue opens this further possibility of a shortage of eigenvectors.

And so there's no second independent eigenvector, x2.

So it's a matrix.

It's a two by two matrix, but with only one

independent eigenvector.

So those are matrices where eigenvectors

**don't give the complete story.**

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#### A deficient matrix

In this course, we will only deal with linear systems of DEs described by complete (not deficient) matrices. Nonetheless, here is an example of a deficient matrix.

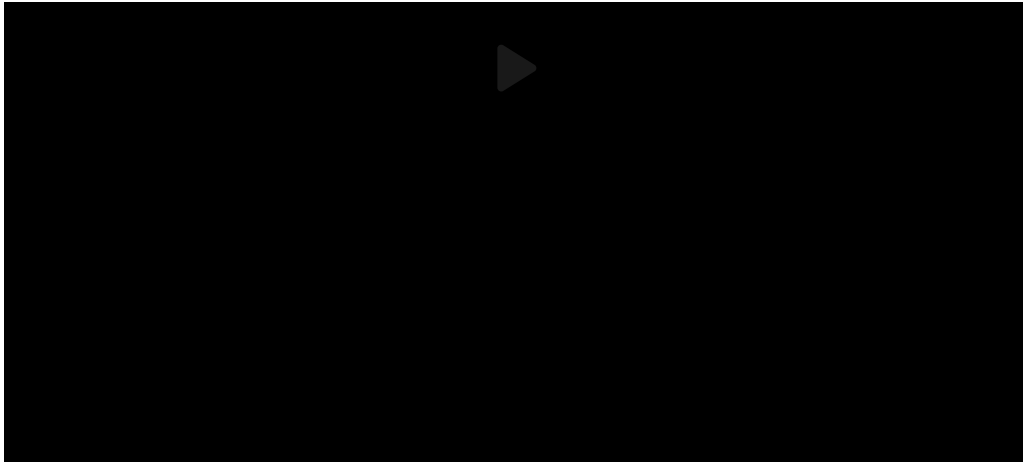
**Example 12.8** The matrix  $\begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$  has characteristic polynomial is  $(\lambda - 1)^2$ , and so its only eigenvalue is **1**, with

## Optional: the spectral theorem

And it's a remarkable fact that you can prove that without--

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All the matrices that you deal with are always symmetric.

And therefore, this repeated eigenvalues is not something **you have to worry about finding extra solutions.**

OK.

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The **transpose** of a matrix **A** is defined to be the matrix **A<sup>T</sup>** formed by turning the rows of **A** into the columns of **A<sup>T</sup>**. In other words, the entry in the *i*th row and *j*th column of **A<sup>T</sup>** is the entry in the *j*th row and *i*th column of **A**:

$$(\mathbf{A}^T)_{ij} = \mathbf{A}_{ji}.$$

**Example 12.9**     The transpose of  $\mathbf{A} = \begin{pmatrix} 1 & 1 & 0 \\ 1 & 0 & 1 \end{pmatrix}$  is the matrix

$$\mathbf{A}^T = \begin{pmatrix} 1 & 1 \\ 1 & 0 \\ 0 & 1 \end{pmatrix}.$$

A **symmetric matrix** is a square *n* × *n* matrix **A** that is equal to its transpose.

$$\mathbf{A}^T = \mathbf{A}.$$

In other words, the entry in the *i*th row and *j*th column is equal to the entry in the *j*th row and *i*th column

$$a_{ij} = a_{ji}.$$

**Example 12.10**     The matrix  $\mathbf{A} = \begin{pmatrix} -2 & 1 & 1 \\ 1 & -2 & 1 \\ 1 & 1 & -2 \end{pmatrix}$  is symmetric.

**Spectral Theorem:**    If **A** is a real, symmetric, *n* × *n* matrix, then all of its eigenvalues are complete.

### 12. Dimension of an eigenspace

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